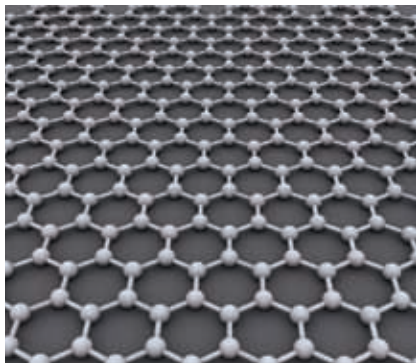


Graphenes

Graphene is an allotrope of carbon, whose structure is most easily visualized as an atomic-scale chicken wire made of carbon atoms and their bonds. The term graphene was coined as a combination of graphite and the suffix -ene by Hanns-Peter Boehm, who described single-layer carbon foils in 1962. The carbon-carbon bond length in graphene is about 0.142 nanos. Graphene sheets stack to form graphite with an interplanar spacing of 0.335 nm, which means that a stack of three million sheets would be only one milli thick. Graphene is the



The structure of Graphene

basic structural element of some carbon allotropes including graphite, charcoal, carbon nanotubes and fullerenes. Graphene won scientists, Andre Geim and Konstantin Novoselov the Nobel Prize in Physics for 2010.

It is speculated that graphene ribbons can be produced by cutting open carbon nanotubes. In one such method multi-walled carbon nanotubes are cut open in solution by action of potassium permanganate and sulfuric acid. In another method, graphene nanoribbons are produced by plasma etching of nanotubes partly embedded in a polymer film.

Potential applications of Graphene:

- ▶ Making components with higher strength to weight ratios
- ▶ Making Transistors that operate at higher frequency
- ▶ Lowering cost of display screens in mobile devices
- ▶ Storing hydrogen for fuel cell powered cars
- ▶ Developing Sensors to diagnose diseases. 